

Abdullah Akgül

Curriculum Vitæ

I build probabilistic algorithms that make reinforcement learning agents learn faster and adapt more reliably. Three of my first-authored algorithms (MOMBO on offline RL, EPPO on non-stationary control, and DAIF on online control) each rank first in sample efficiency on their respective benchmarks, with publications at NeurIPS, ICML, ICLR, and TMLR. Beyond research, I have deployed a deep learning fraud detection system in production at Vakifbank and shipped open-source ML packages (ObjectRL, MOMBO, EPPO) with CI, unit tests, and full API documentation, adopted by the research community. I am seeking research scientist or machine learning engineer roles focused on sequential decision-making, robotics, or autonomous systems.

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 Abdullah Akgül

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EDUCATION

- **Ph.D. in Computer Science** University of Southern Denmark, Denmark Feb 2023 - May 2026
– Thesis: *Probabilistic Reinforcement Learning for Sample-Efficient Control* 
- **M.Sc. in Computer Engineering** Istanbul Technical University, Turkey (GPA: 4.0/4.0) Sep 2020 - Nov 2022
– Thesis: *Memory-based Approaches to Problems in Probabilistic Modeling* 
- **B.Sc. in Computer Engineering** Istanbul Technical University, Turkey (GPA: 3.7/4.0) Sep 2016 - Aug 2020

TECHNICAL SKILLS

Programming Languages: Python, C, C++, JavaScript, HTML/CSS

ML Frameworks & Libraries: PyTorch, JAX, TensorFlow, Keras, Scikit-learn, NumPy, SciPy, Pandas, OpenCV

RL Environments & Benchmarks: MuJoCo/Gymnasium, DeepMind Control Suite (DMC), D4RL, NeoRL, EvoGym, MaMuJoCo, JaxMARL, PettingZoo, PyBullet

Tools & Platforms: Git, Linux, Docker, ROS, MATLAB, LaTeX, Weights & Biases, Flask, Node.js

Languages: English (Fluent), Turkish (Native), Danish (Basic)

Areas of Expertise: Reinforcement Learning (Model-Based, Offline, Multi-Agent), Probabilistic Modeling, Bayesian Inference, Uncertainty Quantification, Deep Learning, Computer Vision, Contextual Bandits, Large Language Models

EXPERIENCE

- **Postdoctoral Researcher** Feb 2026 - Aug 2026
Department of Mathematics and Computer Science, University of Southern Denmark Denmark
– Extending probabilistic RL to multi-agent settings, focusing on non-stationary dynamics in cooperative and competitive environments; co-authored a paper accepted to UAI 2026 on non-stationary linear contextual bandits.
– Contributing to the ADIN Lab open-source ecosystem, including ObjectRL, a pip-installable deep RL package with CI, unit testing, and full API documentation.
- **Ph.D. Researcher** Feb 2023 - Jan 2026
Department of Mathematics and Computer Science, University of Southern Denmark Denmark
– Published at NeurIPS, ICML, ICLR, TMLR, UAI, and L4DC on probabilistic methods for sample-efficient reinforcement learning (ADIN Lab, supervised by Prof. Melih Kandemir); all three first-authored algorithms rank first on their respective benchmarks.
– Built and open-sourced pip-installable Python packages (MOMBO, EPPO, ObjectRL) with unit testing, continuous integration, and full API documentation, adopted by the research community.
– Delivered exercise sessions on core machine learning concepts for undergraduate and graduate courses (~40 students), providing individualized feedback on assignments.
– Mentored 2 MSc students through the full thesis lifecycle, supporting research design and implementation.
- **Research and Teaching Assistant** Mar 2021 - Jan 2023
Artificial Intelligence and Data Engineering, Istanbul Technical University Turkey
– Conducted research in computer vision and uncertainty quantification at the Vision Lab (supervised by Prof. Gözde Ünal); co-authored a paper at ICLR 2022 and contributed to NeurIPS Workshop 2022 and ICLR Workshop 2024.
– Teaching Assistant for Python Programming, Probability & Statistics, and Computer Architecture (100+ students per course).
– Mentored 1 MSc student through the full thesis lifecycle, resulting in a TMLR publication; supervised 3+ BSc students on capstone projects, producing 2 workshop papers at NeurIPS and ICLR.
- **Machine Learning Engineer** Aug 2020 - Sep 2020
R&D and Innovation, Vakifbank Turkey
– Built an end-to-end fraud detection pipeline in TensorFlow 2/Keras: queried a large internal database, automatically extracted and cleaned signature regions from scanned cheque pages, and constructed a labelled training dataset from scratch.
– Trained and deployed a Siamese CNN achieving 95% accuracy on internal data and 88% on the public CEDAR benchmark, generalizing to previously unseen signers without retraining; structured the codebase for handoff to the R&D team.
- **Research Intern** Aug 2019 - Sep 2019
Artificial Intelligence and Robotics Laboratory (AIRLab), Istanbul Technical University Turkey
– Implemented deep reinforcement learning algorithms (DDPG, SAC) for robotic manipulation tasks.
– Developed PyBullet simulation environments with MoveIt integration for inverse kinematics.

•Distributional Active Inference

2026

Akgül, A., Baykal, G., Haußmann, M., ..., Kandemir, M., *International Conference on Machine Learning (ICML)*.   

- Delivered the first model-free integration of Active Inference into distributional RL by parameterizing quantile regression as Bayesian NIG inference, enabling principled uncertainty-driven exploration with no world model; achieved best average rank across all 19 continuous control tasks on three benchmark suites (EvoGym, DMC, DMC Vision), with up to +62% sample efficiency over the next-best baseline.

•Deterministic Uncertainty Propagation for Improved Model-Based Offline Reinforcement Learning

2024

Akgül, A., Haußmann, M., Kandemir, M., *Advances in Neural Information Processing Systems (NeurIPS)*.   

- Diagnosed high-variance Monte Carlo Bellman targets as the primary instability in model-based offline RL and replaced sampling with deterministic progressive moment matching, yielding provably tighter suboptimality bounds and best convergence rate (AULC rank 1.2) across all 12 D4RL settings; advantage is most pronounced on medium-quality datasets (hopper-medium AULC 95.9 vs. 82.2 for MOBILE and 37.0 for MOPO).

•Overcoming Non-stationary Dynamics with Evidential Proximal Policy Optimization

2025

Akgül, A., Baykal, G., Haußmann, M., Kandemir, M., *Transactions on Machine Learning Research (TMLR)*.  

- Demonstrated that an evidential NIG critic simultaneously solves both compounding failures of PPO under non-stationary dynamics (loss of plasticity and undirected exploration) through adaptive gradient scaling and UCB-based advantage estimation; achieved best AULC rank across 10+ non-stationary MuJoCo scenarios (rank 1.5 slippery, 1.7 paralysis) while maintaining higher effective rank and fewer dormant neurons over 15 seeds.

•Evidential Turing Processes

2022

Kandemir, M., Akgül, A., ..., Unal, G., *International Conference on Learning Representations (ICLR)*.   

- Proved the Complete Bayesian Model (CBM) is the minimal structure for total calibration by combining global model uncertainty with local evidential uncertainty; ETP achieves the only simultaneous top performance on NLL (best on all 5 datasets), ECE (2.6–2.7% on image benchmarks), and OOD detection across both text and image benchmarks; designed and conducted all experiments as 2nd author.

•Bridging the Performance-Gap Between Target-Free and Target-Based Reinforcement Learning

2026

Vincent, T., ..., Akgül, A., ..., D'Eramo, C., *International Conference on Learning Representations (ICLR)*.  

- Resolved the target-free vs. target-based RL tradeoff by sharing all backbone parameters and duplicating only the final linear head, reducing memory by up to 49% while matching or exceeding target-based performance across five settings: Atari (+6%), offline Atari (gap 26% → 6%), DMC Hard (full recovery), streaming RL (+10%), and language RL on Wordle (+5%); led the language RL experiments, applying the method to a GPT-2 backbone and demonstrating that the approach transfers to large language models.

•Continual Learning of Multi-modal Dynamics with External Memory

2024

Akgül, A., Unal, G., Kandemir, M., *Learning for Dynamics and Control Conference (L4DC)*.  

- Introduced the first continual learning method for multi-modal dynamical systems: neural episodic memory with a Dirichlet Process prior replaces parameter transfer, enabling automatic mode discovery and catastrophic-forgetting-free adaptation without mode labels; outperforms the VCL-BSSM baseline on 4/5 datasets (NMSE) and 5/5 datasets (NLL), with a 26% NMSE reduction on Character Trajectories.

•Weighted Sequential Bayesian Inference for Non-Stationary Linear Contextual Bandits

2026

Werge, N., Wu, Y.S., Akgül, A., Kandemir, M., *Conference on Uncertainty in Artificial Intelligence (UAI)*. 

- Derived a novel concentration inequality for Weighted Sequential Bayesian posteriors in non-stationary linear contextual bandits, yielding three algorithms (WSB-LinUCB, WSB-RandLinUCB, WSB-LinTS) with improved frequentist regret guarantees over WRLS-based baselines at equivalent computational cost.

•Meta Continual Learning on Graphs with Experience Replay

2023

Unal, A., Akgül, A., Kandemir, M., Unal, G., *Transactions on Machine Learning Research (TMLR)*. 

- Designed a meta-continual learning framework for node classification on graphs combining MAML-style meta-learning with experience replay; improved average accuracy by 10–80% over baselines across three benchmarks (CiteSeer: 51.48%, Arxiv: 31.76%, Reddit: 67.66%) while reducing catastrophic forgetting by up to 78 percentage points.

•ObjectRL: An Object-Oriented Reinforcement Learning Codebase

2025

Baykal, G., Akgül, A., Haußmann, M., Tasdighi, B., Werge, N., Wu, Y.S., Kandemir, M., *arXiv preprint*.  

- Designed ObjectRL, the first fully object-oriented deep RL codebase (9 algorithms, 5 MuJoCo environments) where encapsulation, inheritance, and polymorphism map directly to RL components; extending SAC to a new algorithm requires overriding ~5 lines of code, released as a pip-installable Python package with CI, unit tests, and full API documentation.

•The rest is available in Google Scholar: <https://scholar.google.com/citations?user=FZeaKPoAAAAJ>.

ACADEMIC SERVICE

•**Reviewer:** Advances in Neural Information Processing Systems (NeurIPS) 2025 - Present

•**Reviewer:** European Workshop on Reinforcement Learning (EWRL) 2025 - Present

•**Reviewer:** IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2025 - Present

•**Reviewer:** IEEE Transactions on Neural Networks and Learning Systems (TNNLS) 2022 - 2023